Dipartimento di Ingegneria e Scienza dell'Informazione

– KnowDive Group –

KG 2025 - Trentino Tourism

Document Data:	Reference Persons:
January 24, 2026	Matilde Pereira

Index:

Revision History:

Revision	Date	Author	Description of Changes
1	October 27, 2025	Mayukh Bagchi	Document created

1 Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by iTelos. The KG project documentation plays an important to enhance the reusabiltiy of the resources handled and produced during the process. A clear description of the resources, the process (and sub processes) developed and evaluation at each step of the process provides a clear understanding of the project, thus serving such an information to external readers for the future exploitations of the project's outcomes.

The current document aims to provide a detailed report of the project developed following the iTelos methodology. The report is structured, to describe:

- Section 2: Overall project design choices and introduction.
- Section 3: Definition of the project's purpose and related information gathering.
- Sections 3, 4, 5, 6, 7: The description of the iTelos process phases and their activities, divided by knowledge and data layer activities, as well as the evaluation of the resources produced in terms of fit for the chosen purpose.
- Section 8: The description of the metadata produced for all (and all kind of) the resources handled and generated by the iTelos process, while executing the project.
- Section 9: Conclusion and open issues summary.

2 Project Design

This section describes the overall design of the project and the methodological choices adopted to implement the iTelos framework. The goal of this section is to clarify the project scope, define the expected inputs and outputs, and outline the constraints that guided the modeling and implementation decisions.

2.1 Project Scope and Assumptions

The project focuses on the construction of a Knowledge Graph supporting tourism-related analysis in the Trentino region. Given the available datasets and the scope of a single-student project, the domain of interest is restricted to cultural attractions, natural attractions, accommodation facilities, and their spatial organization at the city level.

The project assumes that:

- cultural heritage entities can be represented at a coarse-grained level (e.g., museums), even when more specific categories exist in reality;
- spatial integration across datasets can be effectively achieved through shared City entities;

- accessibility information can be represented through aggregated indicators rather than fully instantiated auxiliary entities.

These assumptions allow the project to remain coherent with the defined purpose while maintaining a manageable level of complexity.

2.2 Methodological Framework

The project follows the iTelos methodology, instantiated in a lightweight manner consistent with the project scale and available resources.

The methodology is applied as follows:

- the Purpose Definition phase identifies user needs and competency questions;
- the Information Gathering phase analyzes available datasets and reference resources;
- the Language Definition phase stabilizes domain terminology through a Language Teleontology (LTLO);
- the Knowledge Definition phase defines a Teleology aligned with the project purpose;
- the Entity Definition phase instantiates the Knowledge Graph using data mappings;
- the Evaluation phase assesses the resulting Knowledge Graph using structural metrics and query execution.

This phased approach ensures traceability between the initial requirements and the final Knowledge Graph.

2.3 Inputs and Outputs

The main inputs to the project are:

- three semi-formal CSV datasets describing museums, natural attractions, and hotels in the Trentino region;
- reused and adapted ontology resources supporting language and knowledge definition.

The main outputs of the project are:

- a Language Teleontology (LTLO);
- a Knowledge Teleontology and project-specific Teleology (KTLO/TLO);
- an Entity Graph Model (R2RML mappings);
- a populated Knowledge Graph serialized in RDF (Turtle format);
- SPARQL queries and query results validating the project objectives;
- this project report documenting the entire process.

2.4 Design Constraints

Several constraints influenced the project design.

First, the project was developed by a single student within a limited time frame, motivating the adoption of a reduced but coherent modeling scope. Second, the available datasets do not provide uniform coverage of all tourism-related aspects, limiting the instantiation of certain entities and relationships. Finally, tool-related constraints (e.g., the operational requirements of KARMA) influenced the adoption of a flattened Teleology for the Entity Definition phase.

These constraints were explicitly considered during the design phase and informed the modeling and abstraction choices discussed throughout the report.

3 Purpose Definition

The iTelos methodology provides a structured and iterative approach aimed at reducing the effort in building Knowledge Graphs (KGs) for a well-defined purpose expressed by the final user.

This section describes the first methodological phase, Purpose Definition, applied to the topic "Territory and Tourism Facilities", specified in the domain of cultural heritage.

3.1 Informal Purpose

The goal of this project is to build a Knowledge Graph (KG) that stores, organizes, and links information about cultural and heritage sites in the Trentino region, with a particular focus on museums, art galleries, castles, archaeological and geological sites.

The KG aims to serve as a central hub for tourists, cultural heritage, experts, and tourism operators, facilitating access to and analysis of Trentino's rich historical and cultural assets. The project emphasizes connections between sites, their geographical context, and related facilities, enabling users to perform semantic queries such as:

- "Which castles are located near Lake Garda?"
- "What museums in Trento offer guided tours?"
- "Are there archaeological sites in Val di Non with visitor parking available?"

The KG is intended to promote data integration, interoperability, and usability for both exploration and analytical tasks. Due to data availability constraints, more specific cultural heritage categories are abstracted into broader entity types.

3.2 Domain of Interest (DoI)

The Domain of Interest (DoI) for this project is the Trentino region in Northern Italy, characterized by its diverse cultural and natural heritage.

The region includes a wide range of historical buildings, museums, archaeological areas, and geological formations, distributed throughout valleys, towns, and protected areas.

Key aspects of the domain include the following.

- Cultural Heritage and History: Castles, museums, art galleries, and archaeological sites showcasing Trentino's historical development and artistic legacy.
- Natural and Geological Heritage: Unique landscapes and formations such as caves, fossil sites, and mountain passes of geological interest.
- Tourism and Visitor Facilities: Services such as tourist information centers, guided tours, rest areas, and accessibility features that enhance visitor experience.

3.3 Scenarios Definition

The following scenarios describe possible real-world uses of the Knowledge Graph, addressing different user goals and perspectives.

- Eric Miller, a 22-year-old tourist who is visiting Trento for a weekend, wants to explore nearby museums and art galleries. He hopes to find which ones are open on Sunday and located within walking distance of his hotel.
- Luca Bianchi, a regional tourism officer, wants to analyze the distribution of castles in Val di Non and identify which of them have visitor facilities, such as info points, parking, or guided tours.
- Drª. Maria Esposito, a cultural heritage researcher, studies archaeological and geological sites in Trentino. She uses the KG to find areas where archaeological remains are near notable geological formations.
- Elisa Becker, a travel planner, is preparing a cultural itinerary for a group of visitors. She needs to know which towns offer both cultural attractions and nearby accommodations.
- Adam Taylor, a writer, is coming to Trentino for inspiration for his future book. He wants to know the relationships the Lavini di Marco sight has with other places around the area.

3.4 Personas

- Eric Miller - 22 years old, from Melbourne, passionate about culture and art, often visits places around the world for week long or weekend getaways with his partner, and uses online platforms to plan. He wants to find and compare museums and galleries efficiently.
- Luca Bianchi - 41 years old, works for the Trentino Tourism Board. His goal is to manage and analyze data on cultural sites to improve regional tourism strategies.
- Drª. Maria Esposito - 35 years old, researcher in archaeology at the University of Madrid. She focuses on the relationship between cultural heritage and natural landscapes, has a background in the study of the area of Cáceres, Spain. Recently married to a Trentino doctor, given having to move to the region, she felt compelled to explore the historical heritage of the Trentino area.

- Elisa Becker - 32 years old, travel planner for a German agency. She designs cultural tours in central Europe and needs access to accurate and structured information on tourist facilities.
- Adam Taylor - 39 years old, British writer. After finding himself in writers block for the last 3 years after his World Best Seller, he is trying to find inspiration in world sights around the world related to writing. Vehemently believing that seeing Lavini di Marco in person will let him understand Dante's Inferno, Chapter XII better, he also wants to know how the sight relates to others around it and also where to stay for his trip to Rovereto.

3.5 Competency Questions (CQs)

These questions reflect the information needs of the defined personas and guide the data modeling and source selection in later phases.

- CQ1 - What museums or art galleries are located in the Trento municipality?
- CQ2 - Which castles or archaeological sites are near known natural landmarks such as lakes or mountains
- CQ3 - What geological points of interest exist in the general Trentino area?
- CQ4 - Which cultural heritage sites provide visitor facilities such as info points, parking, or guided tours?
- CQ5 - What cultural attractions are thematically or spatially connected(e.g., "castle route", "archaeological network")?
- CQ6 - How many cultural sites of each type exist per municipality?
- CQ7 - Which sites are part of official heritage inventories such as UNESCO or regional heritage lists?

3.6 Concepts Identification

This step identifies the key entity types (Etypes) and their properties, derived from the Competency Questions and aligned with iTelos' popularity categories (common, core, contextual).

3.7 ER model definition

In Figure 1 is represented the ER model for the KG.
Common entities:

- Place (geographical context)
- CulturalSite (main subject of queries)

Core entities:

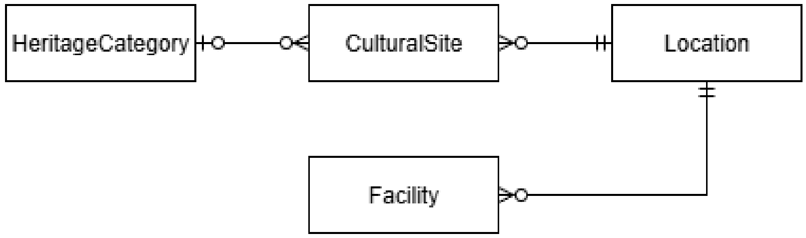

Figure 1: ER Model

- HeritageCategory (classification of cultural sites)

Contextual entities:

- Facility (supporting elements around sites)

3.8 Information Gathering

This subsection describes the Information Gathering phase of the iTelos methodology. The objective of this phase is to identify, collect, and analyze the knowledge and data resources required to support the construction of the Knowledge Graph, in alignment with the purpose and competency questions defined in the previous phase.

The Information Gathering activity involved the analysis of resources with different levels of formality, including knowledge-level resources (teleontologies and reference fragments) and data-level resources (datasets). The selection of resources was guided by relevance to the project purpose, data availability, and feasibility of integration within the scope of the project.

3.8.1 Knowledge Resources

The knowledge resources provided for the project include a Knowledge Teleontology (KTLO), a Language Teleontology (LTLO), and a set of reference ontology fragments. These resources were analyzed to identify reusable concepts and relationships relevant to the domain of cultural heritage, tourism, and geographical entities.

The KTLO was used as a reference backbone for defining upper-level and domain-level entity types and properties during the Knowledge Definition phase. The LTLO was considered to support the alignment of natural language terms extracted from the datasets and competency questions, ensuring consistency and reducing ambiguity in terminology.

Reference ontology fragments related to places, points of interest, and facilities were evaluated and selectively reused. Only fragments directly relevant to the project purpose were considered, while others were excluded to maintain a focused and manageable knowledge structure.

3.8.2 Museum Dataset

The Museum dataset is a semi-formal data resource provided in CSV format. It contains information about museums located in the Trentino region and represents the primary data source for modeling cultural heritage entities in the Knowledge Graph.

The dataset includes the following main attributes:

- id
- Name
- Latitude
- Longitude
- City
- Address
- Phone_number

This dataset provides both identifying and spatial information, enabling the representation of museums as cultural points of interest and their integration with geographical entities such as cities. The presence of latitude and longitude values allows spatial reasoning and proximity-based analysis.

No major data-cleaning issues were identified, aside from minor inconsistencies in formatting, which were addressed during the data preparation process.

3.8.3 Natural Attraction Dataset

The Natural Attractions dataset is a semi-formal CSV dataset containing information about natural sites and attractions in the Trentino region. This dataset was selected to represent natural points of interest and to complement the cultural focus provided by the museum dataset.

The main attributes of the dataset are:

- id
- Name
- Latitude
- Longitude
- n_bus

The attributes Latitude and Longitude support spatial representation, while the attribute n_bus provides aggregated information about nearby public transportation accessibility. The attribute n_bus refers to identifiers of nearby bus stops and is associated with a dedicated semi-formal dataset describing bus stop entities. However, in order to maintain a lightweight and focused knowledge structure, bus stops were not fully instantiated as standalone entities during the Entity Definition phase. Instead, the n_bus attribute was used to preserve references to bus stop identifiers, allowing the representation of public transportation accessibility without introducing additional complexity.

This dataset contributes to answering competency questions related to the spatial distribution of natural attractions and their accessibility.

3.8.4 Hotel Dataset

The Hotel dataset is a semi-formal CSV dataset containing information about hotel facilities in the Trentino region. It was selected to represent accommodation services and to support scenarios involving tourists seeking lodging near cultural or natural attractions.

The dataset includes the following attributes:

- id
- Name
- Latitude
- Longitude
- City
- Address
- Phone_number
- n_barpub

The dataset provides both spatial and contact information, enabling the integration of accommodation entities with other points of interest through shared geographical context (City, coordinates).

The attribute n_barpub refers to nearby bar or pub services. However, since a dedicated Bar/Pub dataset was not available, this attribute was treated as an aggregated indicator of nearby services rather than as a reference to individual entities.

3.8.5 Data Selection and Exclusions

Although additional data sources were provided for the project, including OpenStreetMap data, GTFS resources, and schema.org references, these resources were not integrated into the final Knowledge Graph.

The decision to exclude these resources was motivated by the project's scope and purpose. Transportation scheduling and detailed service modeling were not central to the defined competency questions, and the available CSV datasets already provided sufficient spatial and contextual information to satisfy the main project objectives.

This selective approach allowed the project to focus on data integration quality rather than data quantity, ensuring coherence between the purpose definition and the available resources.

A dataset describing bus stop entities was evaluated and partially reused through identifier references, without full entity instantiation, as public transportation scheduling was not central to the project's competency questions.

3.8.6 Information Gathering Report

The Information Gathering phase highlighted the strong dependency between data availability and knowledge modeling choices. While an initial broader scope was considered, the final selection of three datasets (museums, natural attractions, and hotels) was driven by relevance, completeness, and integration feasibility.

Some limitations were identified, particularly regarding the absence of detailed datasets for auxiliary services such as bar/pub facilities and public transportation entities. These limitations influenced later modeling decisions, leading to the use of aggregated accessibility indicators instead of fully instantiated service entities.

Overall, the Information Gathering phase provided a sufficient and coherent set of resources to support the subsequent Language Definition, Knowledge Definition, and Entity Definition phases, while maintaining a manageable project scope aligned with the defined purpose.

3.9 Purpose Definition Evaluation

Table 1: Evolution of entities and competency questions from initial purpose definition to final project scope.

Aspect	Initial	Final
ER entities	4	4
Competency questions	6	3
Scenarios	broad tourism	cultural, accommodation

During the Purpose Definition phase, a broad set of scenarios and competency questions was initially identified, covering multiple aspects of tourism in the Trentino region, including cultural heritage classification, visitor facilities, and transportation infrastructure.

As the project progressed, the initial purpose was refined to align with the structure and coverage of the available datasets. In particular, competency questions requiring fine-grained cultural heritage categories, detailed transportation modeling, or auxiliary facilities were discarded or abstracted into higher-level concepts.

This refinement resulted in a reduced but coherent set of competency questions focused on museums, natural attractions, hotels, and their spatial organization at the city level.

4 Language Definition

This section describes the Language Definition phase of the iTelos methodology. The objective of this phase is to define and stabilize the language used to represent the domain knowledge, by identifying relevant terms and aligning them with the provided Language Teleontology (LTLO). This process ensures clarity, consistency, and reusability of terminology across the Knowledge Graph.

The Language Definition phase focuses on defining the vocabulary for entity types, data properties, and object properties that will be used in the subsequent Knowledge Definition and Entity Definition phases.

4.1 Concept Identification

The first activity of the Language Definition phase consisted in identifying the key language terms required to represent the information needed to satisfy the project purpose and competency questions. These terms were derived from the scenarios, competency questions, and attributes of the selected datasets.

The identified terms were classified into three main categories:

- entity type terms (etypes),
- data property terms,
- object property terms.

Special attention was paid to terms recurring across multiple datasets, as these represent shared concepts and support data integration.

4.1.1 Entity Type Terms

The following entity type terms were identified as relevant for the project:

- Place
- City
- PointOfInterest
- Museum
- NaturalAttraction
- Hotel
- BusStop
- BarPub

Some of these terms represent abstract or upper-level concepts (e.g., Place, PointOfInterest, Accommodation), while others represent domain-specific concepts directly instantiated from the datasets (e.g., Museum, NaturalAttraction, Hotel).

The terms BusStop and BarPub were identified as contextual concepts. Although not fully instantiated during the Entity Definition phase, they were included at the language level to support accessibility-related reasoning and future extensions.

4.1.2 Data Property Terms

Data property terms were identified primarily from the attributes present in the selected CSV datasets. The following data property terms were defined:

- identifier

- hasName
- hasLatitude
- hasLongitude
- hasAddress
- hasPhoneNumber
- hasNearbyBusStop
- hasNearbyBarPub

The properties hasNearbyBusStop and hasNearbyBarPub represent references to identifiers of related contextual entities, rather than fully instantiated object properties.

4.1.3 Object Property Terms

A limited set of object property terms was identified, reflecting the lightweight relational structure required by the project purpose:

- locatedIn
- relatedTo

The object property locatedIn is used to express spatial containment between points of interest or accommodations and cities. The object property relatedTo is used to represent conceptual proximity between hotels and nearby services (e.g., bar or pub facilities).

4.2 Language Teleontology Alignment

Once the relevant language terms were identified, they were aligned with the provided Language Teleontology whenever possible. Alignment was performed by associating each term with an appropriate parent concept in the LTLO, based on semantic similarity and intended usage.

Upper-level terms such as Place and City were directly aligned with existing LTLO concepts related to geographical entities. Similarly, terms such as Museum and Hotel were aligned with LTLO concepts representing cultural institutions and accommodation facilities.

However, some domain-specific terms required specialization beyond the available LTLO definitions. In particular, the terms NaturalAttraction, BusStop, and BarPub did not have direct equivalents with the required level of specificity. These terms were therefore introduced as extensions to the LTLO, with informal definitions consistent with their usage in the datasets.

This alignment process resulted in a partial enrichment of the Language Teleontology.

4.3 Dataset Filtering and Language Alignement

The Dataset Filtering activity aimed to ensure that the data-level resources were compatible with the stabilized language defined in this phase. In this project, the selected datasets were already well aligned with the identified language terms, and no additional filtering was required beyond minor normalization of attribute names.

Attributes referring to accessibility-related concepts (e.g., bus stops and bar/pub services) were preserved as identifier-based references, in order to maintain consistency with the language-level modeling choices and to avoid introducing unnecessary complexity.

4.4 Language Definition Discussion and Evaluation

The Language Definition phase resulted in a consistent and purpose-driven vocabulary that supports the representation of cultural and natural attractions together with accommodation facilities and accessibility indicators. The adoption of a lightweight language model, combined with selective enrichment of the Language Teleontology, ensured clarity and reusability while remaining aligned with the scope and constraints of the project.

The stabilized language defined in this phase provides the foundation for the formal knowledge structures defined in the subsequent Knowledge Definition phase.

Following the refinement of the project purpose, the language resources were pruned to retain only those terms required to support the final competency questions. In total, eight entity types were defined at the language and teleological level, of which four were instantiated in the final Knowledge Graph (Museum, NaturalAttraction, Hotel, and City) were selected as aligned and instantiated in the Knowledge Graph.

Additional entity types, such as BusStop and BarPub, were retained at the language level to support aggregated accessibility indicators but were not instantiated as standalone entities. Higher-level abstractions not required by the final use cases were discarded.

This pruning process ensured alignment between language resources, the Teleology, and the data layer, avoiding unnecessary complexity.

5 Knowledge Definition

The Knowledge Definition phase represents a central step in the iTelos methodology, as it aims to formalize the knowledge structures required to represent the information identified in the previous phases. In this phase, the Teleology (TLO) is defined by composing and specializing concepts from the provided Knowledge Teleontology and reference fragments, while ensuring alignment with the project purpose and the selected datasets.

The outcome of this phase is a coherent teleological structure that supports the integration of multiple datasets and enables the representation of cultural and natural attractions together with tourism-related facilities.

5.1 Teleology Definition

The Teleology defined for this project focuses on representing points of interest in the Trentino region, with particular attention to cultural sites (museums), natural attractions, and accommodation facilities (hotels). The Teleology is designed to support spatial reasoning and accessibility analysis, allowing users to explore relationships between attractions and nearby accommodations, as well as basic accessibility indicators.

The Teleology structure is centered around the concept of PointOfInterest, which represents any location of potential interest to visitors, and Accommodation, which represents facilities offering lodging services. These concepts provide a shared semantic backbone that enables the integration of heterogeneous datasets.

5.2 Core Entity Types

Based on the competency questions and the analysis of the selected datasets, the following entity types were identified.

Common Entity Types

- Place, representing geographical entities.
- PointOfInterest, representing locations of touristic interest.
- Accommodation, representing lodging facilities.

These entity types are considered common, as they provide shared properties and enable reuse across multiple datasets.

Domain-Specific Entity Types

- Museum, a specialization of PointOfInterest representing cultural institutions open to visitors.
- NaturalAttraction, a specialization of PointOfInterest representing natural sites of environmental or geological interest.
- Hotel, a specialization of Accommodation representing hotel facilities.
- City, a specialization of Place representing municipalities within the Trentino region.

Additionally, the entity type BarPub was identified as a relevant contextual concept to model nearby services related to hotels. However, due to the absence of a dedicated dataset, this entity type is included in the Teleology but is not instantiated in the Entity Definition phase. The entity type BusStop was considered at the language level to support accessibility-related reasoning; however, it was not promoted to a fully instantiated entity type in the Teleology, as transportation infrastructure modeling was outside the project scope.

5.3 Properties Definition

5.3.1 Object Properties

The following object properties were defined to represent relationships between entity types:

- locatedIn, relating a PointOfInterest or an Accommodation to a City, representing spatial containment.
- relatedTo, relating a Hotel to a BarPub, representing conceptual proximity to nearby services.

The relatedTo property is included at the Teleology level to capture potential service relationships, even though it is not fully instantiated due to data availability limitations.

5.3.2 Data Properties

Data properties were defined based on the attributes available in the selected datasets. These properties enable the representation of identifying, spatial, and accessibility-related information.

Common data properties include:

- identifier
- hasName
- hasLatitude
- hasLongitude

Additional data properties include:

- hasAddress
- hasPhoneNumber
- hasNearbyBusStopsCount
- hasNearbyBarPubCount

Accessibility-related properties were modeled as aggregated data properties rather than object properties, in order to represent information derived from the datasets without requiring additional entity instantiation.

5.4 Alignment with the Knowledge Teleontology

Whenever possible, the entity types and properties defined in the Teleology were aligned with corresponding concepts in the provided Knowledge Teleontology. Upper-level concepts such as Place, PointOfInterest, and Accommodation were aligned with existing KTLO concepts related to spatial entities and facilities.

Domain-specific concepts such as Museum, NaturalAttraction, and Hotel required specialization beyond the available KTLO definitions and were therefore introduced as extensions to better support the project purpose. This process resulted in a partial enrichment of the Knowledge Teleontology.

5.5 Consequences for Competency Questions

The Teleology defined in this phase supports the majority of the competency questions defined during the Purpose Definition phase. However, some competency questions related to specific service entities (e.g., detailed information about bar/pub facilities) can only be partially addressed due to the lack of dedicated datasets.

Some competency questions refer to specific categories of cultural heritage sites, such as castles, towers, or archaeological sites. While no dedicated datasets were available for these categories, several entities contained in the Museum dataset represent sites that also belong to these categories from a cultural or historical perspective. In this project, such entities were modeled as Museum instances, reflecting their primary function as visitor-accessible cultural institutions. This modeling choice allows partial coverage of the corresponding competency questions, while avoiding the introduction of additional entity types not explicitly supported by the available data.

These limitations highlight the dependency between data availability and knowledge modeling and justify the design choices adopted during this phase.

5.6 Discussion

The Teleology provides a structured and purpose-driven representation of the domain, enabling the integration of cultural, natural, and accommodation-related information within a single Knowledge Graph. The design choices adopted in this phase balance semantic expressiveness and data availability, ensuring that all modeled concepts can be meaningfully instantiated in the subsequent Entity Definition phase.

6 Entity Definition

This section describes the Entity Definition phase of the iTelos methodology. The objective of this phase is to merge the knowledge layer, defined during the Knowledge Definition phase, with the data layer resources collected and analyzed during the Information Gathering phase, resulting in the construction of the final Knowledge Graph (KG).

The Entity Definition phase addresses heterogeneity at the data value level by identifying, matching, and mapping entities from different datasets to the teleological structure defined previously.

6.1 Overview and Objectives

The input of the Entity Definition phase consists of:

- the Teleology (TLO) defined in the Knowledge Definition phase;
- the cleaned and aligned semi-formal datasets describing museums, natural attractions, hotels, and bus stops.

The output of this phase is a structured Knowledge Graph in which data values are instantiated as entities and linked according to the defined teleology.

The main objectives of this phase are:

- to identify entities represented in the datasets;
- to ensure consistent and unique identification of entities;
- to map data values to the corresponding entity types and properties defined in the Teleology.

6.2 Entity Matching

Entity matching aims to resolve potential ambiguities arising from the presence of similar or overlapping real-world entities across multiple datasets.

In this project, each primary entity type (Museum, NaturalAttraction, Hotel) is represented by a single dataset. Therefore, no complex cross-dataset entity matching was required at the value level.

However, spatial entities such as cities appear across multiple datasets. These entities were implicitly matched by normalizing city names and treating them as shared instances of the City entity type. This approach ensured consistency when linking museums and hotels to their respective cities.

Bus stop identifiers appearing in the Natural Attractions dataset were not matched to fully instantiated BusStop entities, but were preserved as references to maintain accessibility-related information.

6.3 Entity Identification

Entity identification focuses on ensuring that each real-world entity is uniquely and consistently represented in the Knowledge Graph.

For each dataset, the provided id attribute was used as the primary identifier for entity instantiation. Identifiers were combined with the entity type name when necessary to ensure global uniqueness within the Knowledge Graph.

For example:

- Museum entities were identified using the identifiers provided in the museum dataset;
- NaturalAttraction entities were identified using the identifiers provided in the natural attractions dataset;
- Hotel entities were identified using the identifiers provided in the hotel dataset.

City entities were identified using normalized city names, allowing multiple entities to reference the same City instance without duplication.

Bus stop identifiers and bar/pub references were maintained as literal identifiers rather than as fully instantiated entities, in line with the lightweight modeling choices adopted in the Knowledge Definition phase.

6.4 Entity Mapping

Entity mapping consists in associating data values from the datasets with the corresponding entity types, data properties, and object properties defined in the Teleology.

For each dataset, entities were instantiated at the level of leaf classes in the Teleology, namely Museum, NaturalAttraction, and Hotel. Data properties such as hasName, hasLatitude, hasLongitude, hasAddress, and hasPhoneNumber were populated directly from the corresponding dataset attributes.

Spatial relationships were modeled using the object property locatedIn, linking Museum and Hotel entities to City entities. Accessibility-related information was modeled using data properties referring to contextual identifiers, such as hasNearbyBusStop and hasNearbyBarPub, without instantiating additional entities.

This mapping approach ensured that all instantiated entities were directly usable for answering the defined competency questions, while avoiding unnecessary complexity in the Knowledge Graph structure.

6.5 Phase Outcomes

The Entity Definition phase resulted in the construction of a Knowledge Graph integrating multiple datasets into a unified semantic structure.

The final Knowledge Graph includes:

- entities representing museums, natural attractions, hotels, and cities;
- spatial relationships linking points of interest and accommodations to their respective cities;
- accessibility-related attributes capturing proximity to bus stops and nearby services.

The resulting Knowledge Graph is coherent, consistent with the defined Teleology, and suitable for addressing the project's competency questions.

6.6 Decisions and Reflections

Several design decisions were taken during the Entity Definition phase to balance expressiveness and feasibility.

First, entity instantiation was limited to leaf classes in the Teleology. Abstract and upper-level classes such as PointOfInterest and Accommodation were used to organize the knowledge structure but were not directly instantiated, as no competency questions required reasoning at that level.

Second, contextual entities such as BusStop and BarPub were not fully instantiated due to the lack of dedicated or sufficiently detailed datasets. Instead, references to these entities were preserved as identifier-based data properties. This decision allowed the representation of accessibility-related information while maintaining a lightweight and manageable Knowledge Graph.

City entities are identified by their URI values derived from the dataset, without explicit name literals, as city names are directly encoded in the entity identifiers. This choice is sufficient to support spatial integration across datasets. They were shared across datasets to support integration and avoid duplication, enabling consistent spatial reasoning across museums and hotels.

Overall, the Entity Definition phase successfully merged the knowledge and data layers, producing a Knowledge Graph aligned with the project purpose and constrained by realistic data availability.

7 Evaluation

This section presents the evaluation of the final Knowledge Graph produced through the application of the iTelos methodology. The evaluation follows the structure prescribed by the project template and focuses on: (i) Knowledge Graph statistics, (ii) knowledge layer evaluation, (iii) data layer evaluation, and (iv) query execution. Particular attention is paid to the evolution of the project scope across phases and to the consistency between the defined purpose, the teleology, and the instantiated data.

7.1 Knowledge Graph Information Statistics

The final Knowledge Graph instantiates a subset of the entity types defined during the Language and Knowledge Definition phases. While the Teleology includes a broader set of conceptual entity types, only those required to satisfy the final competency questions are instantiated.

The instantiated etypes are:

- Museum
- NaturalAttraction
- Hotel
- City

The Knowledge Graph statistics are summarized as follows:

- Number of instantiated etypes: 4
- Number of object properties: 1 (locatedIn)
- Number of data properties: 7

The distribution of entities per etype is as shown in 2.

These values were obtained through SPARQL count queries executed in GraphDB.

Table 2: Distribution of instantiated entities by entity type in the final Knowledge Graph.

EType	Number of entities
Museum	41
NaturalAttraction	46
Hotel	295
City	54

7.2 Knowledge Layer Evaluation

The evaluation of the knowledge layer assesses whether the Teleology adequately supports the competency questions defined for the project.

The final set of competency questions focuses on the identification of cultural attractions, natural attractions, accommodation facilities, and their spatial organization at the city level. Accordingly, the set of required entity types is:

$$CQE = \{Museum, NaturalAttraction, Hotel, City\}$$

All required entity types are explicitly defined in the Teleology and instantiated in the Knowledge Graph. The entity type coverage with respect to the competency questions is therefore:

$$CovE(CQE) = \frac{|CQE \cap TE|}{CQE} = \frac{4}{4} = 1.0$$

More specialized concepts initially considered during the Purpose Definition phase (e.g., castles, archaeological sites, transportation facilities) are not represented as distinct entity types. These concepts were either abstracted into broader categories (e.g., Museum) or excluded due to the lack of suitable datasets. This pruning process reflects a deliberate alignment between the Teleology and the available data rather than a limitation of the modeling approach.

7.3 Data Layer Evaluation

The data layer evaluation focuses on entity connectivity and the structure of relationships instantiated in the Knowledge Graph.

7.3.1 Entity Connectivity

Entity connectivity is assessed through a connectivity matrix reporting the number of object-property links between pairs of entity types. The table below reports the connectivity matrix for the final Knowledge Graph.

The matrix highlights a star-shaped topology centered on City entities, which act as integration hubs across datasets. Museums and hotels are connected to cities via the locatedIn object property, while natural attractions remain unlinked at the city level due to the absence of location attributes in the corresponding dataset.

Table 3: Connectivity matrix showing the number of object-property links between entity types in the final Knowledge Graph.

From To	Museum	NaturalAttraction	Hotel	City
Museum	-	0	0	40
NaturalAttraction	0	-	0	0
Hotel	0	0	-	292
City	40	0	292	-

This structure directly reflects the schemas of the input datasets and does not introduce artificial relationships unsupported by the data.

7.3.2 Property Connectivity

The Knowledge Graph includes seven data properties describing entity identification, spatial attributes, and accessibility indicators. The average number of properties per Museum entity is 5.88, indicating a balanced level of description.

The limited number of object properties and the absence of multi-hop relationships are consistent with the project's scope and the intended use cases.

7.4 Query Execution

To validate the suitability of the Knowledge Graph with respect to the project purpose, a subset of competency questions was translated into SPARQL queries and executed in GraphDB.

Out of the initially defined competency questions, three were selected for implementation as SPARQL queries, focusing on:

- museums located in a given city,
- hotels located in the same city as museums,
- natural attractions together with accessibility indicators.

All executed queries returned non-empty and meaningful results, confirming that the Knowledge Graph satisfies the final project purpose within the constraints imposed by data availability.

7.5 Summary of the Evaluation

The evaluation confirms that the final Knowledge Graph is consistent with the defined Teleology and effectively supports the selected competency questions. The observed limitations, particularly regarding entity connectivity and concept granularity, are attributable to the structure of the available datasets and were explicitly addressed through scope reduction and abstraction during the modeling process.

8 Metadata Definition

This section describes the metadata defined for the resources produced during the execution of the iTelos methodology. Metadata were defined with the objective of supporting documentation, reuse, and evaluation of the project outcomes, in accordance with the guidelines provided for the course.

Metadata are provided as structured spreadsheet files and are included in the submission repository.

8.1 Project Metadata

Project-level metadata were defined to describe the overall characteristics of the Knowledge Graph project, including authorship, licensing, scope, and institutional context.

The project metadata are provided in the file project_metadata.xlsx and include information such as:

- project title and category,
- authorship and contact information,
- license,
- keywords and tags,
- publication date.

These metadata describe the project as a whole and are independent of the specific datasets or ontologies used.

8.2 Language Resources Metadata

Metadata were defined for the Language Teleontology (LTLO) produced during the Language Definition phase.

The language resource metadata are provided in the file language_metadata.xlsx and describe:

- the purpose and domain of the language resource,
- ownership and responsibility,
- versioning and creation date,
- reference ontologies and conceptual reuse,
- intended usage context.

These metadata support the interpretation and potential reuse of the language resources developed for the project.

8.3 Knowledge Resources Metadata

No standalone knowledge resource metadata were produced for the datasets used in this project.

The datasets employed during the Information Gathering phase were obtained from a teaching-oriented GitHub repository provided as part of the course materials. The repository does not specify formal data provenance, licensing information, or original publishers for the individual CSV files. As a result, defining dataset-level metadata would require assumptions not supported by the available documentation.

For this reason, dataset metadata are described qualitatively in the Information Gathering section of the report rather than being formalized in dedicated metadata spreadsheets.

8.4 Summary

In summary, metadata were explicitly defined for:

- the project itself, and
- the language resources produced during the iTelos process.

Dataset metadata were intentionally not formalized due to the lack of reliable provenance information. This choice reflects a conservative and transparent approach consistent with the available documentation and the educational context of the project.

9 Open Issues

This section summarizes the main limitations and open issues identified at the end of the project, together with possible directions for future work.

9.1 Project Scheduling and Execution

The project followed the planned sequence of the iTelos methodology phases. Minor delays were encountered during the Entity Definition phase due to the learning curve associated with ontology modeling tools (e.g., Protégé and KARMA). However, these delays did not compromise the completion of the project or the quality of the final Knowledge Graph.

Overall, the project respected the expected scheduling constraints for a single-student project.

9.2 Purpose Satisfaction

The final Knowledge Graph satisfies the core objectives defined in the Purpose Definition phase. In particular, it supports:

- the identification of cultural and natural attractions in the Trentino region,

- the integration of points of interest with accommodation facilities,
- spatial reasoning through shared City entities,
- basic accessibility analysis through aggregated indicators.

Some aspects of the original purpose are only partially satisfied. Competency questions referring to:

- specific categories of cultural heritage sites (e.g., castles, archaeological sites),
- detailed visitor facilities,
- explicit transportation infrastructure modeling,

could not be fully addressed due to limitations in the available datasets. These limitations were identified early during the Information Gathering phase and influenced subsequent modeling decisions.

9.3 Remaining Open Issues

The most relevant open issues at the end of the project are the following:

1. Lack of fine-grained cultural heritage classification - Some entities modeled as Museum instances also represent castles, towers, or archaeological sites. The absence of dedicated datasets prevented the introduction of more specialized entity types.
2. Partial modeling of accessibility-related entities - Bus stops and bar/pub facilities were not fully instantiated as entities, limiting more detailed reasoning about transportation and service proximity.
3. Limited reuse of external reference ontologies - Although reference ontologies were analyzed, only a small subset was reused to maintain a focused scope. Broader alignment could improve interoperability in future extensions.
4. The absence of a direct link between natural attractions and cities limits certain spatial queries and reflects the structure of the available dataset.

9.4 Future Work

Possible future work include:

- integration of OpenStreetMap or GTFS datasets to explicitly model transportation infrastructure,
- refinement of cultural heritage classifications through additional datasets,
- enrichment of City entities with explicit labels and administrative metadata,
- extension of the Knowledge Graph to support temporal and event-based reasoning.

These extensions were not pursued in the current project to maintain coherence with the defined scope and constraints.